OPMIG-07: Metric & Event Extraction

> **Series:** OPMIG — OpenPipeline Migration | **Notebook:** 7 of 10 | **Created:** December 2025 | **Last Updated:** 10/14/2026

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Learning Objectives

By completing this notebook, you will:

1. Configure metric and event extraction from logs
2. Generate business events for analytics
3. ⭐ **NEW:** Implement SLI metrics using RED methodology (Rate, Errors, Duration)
4. ⭐ **NEW:** Extract business KPIs (revenue, conversion, user journey)
5. ⭐ **NEW:** Apply metric design best practices (cardinality, naming, aggregation)

6. Validate extractions with DQL queries
7. Optimize extraction for cost efficiency

Prerequisites

Requirement	Details
Dynatrace Environment	Dynatrace SaaS with Grail and OpenPipeline access — Managed is not covered by this series
Permissions	`settings:read`, `settings:write` (OpenPipeline configuration)
API Access	`logs.read` for the verification queries; `logs.ingest` for the log shipper. Extraction itself needs no `metrics.ingest` or `events.ingest` scope — it runs inside the pipeline on records already ingested
Knowledge	OPMIG-01 through OPMIG-06; understanding of processing/parsing concepts

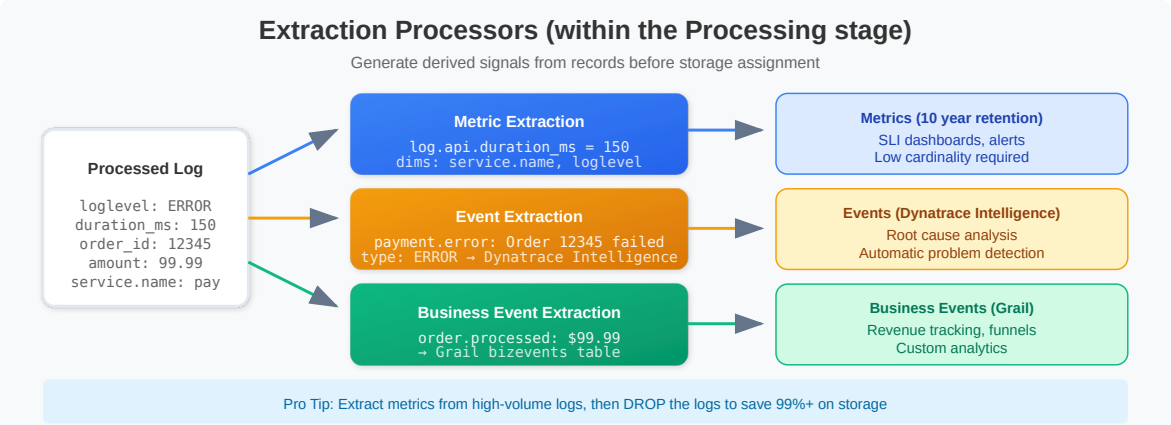
> **Sources:** Processing in OpenPipeline (DT docs) — Metric extraction: *"Extract metrics from the records that match the query."*; custom pipelines: *"OpenPipeline Permissions"* `settings:read`, `settings:write`. **Derived:** extraction acts on records already ingested, so it adds no client-side ingest scope.

Extraction Processors Overview

> **Stage model:** Extraction runs in three dedicated stages after Bucket assignment — **Metric extraction**, **Davis** and **Data extraction** (business events, SDLC events). Smartscape node and edge extraction are their own stages too, straight after Processing. The stage sequence is fixed; OPMIG-02 § *Understanding Processing Order* has the full table.

> **Sources:** Processing in OpenPipeline (DT docs) — *"The sequence of stages is fixed for all pipelines and cannot be modified."*

The extraction processors generate derived signals (metrics, events, business events, Smartscape topology) from records before they reach Storage:



> **Span sampling note:** When extracting metrics from spans, use the **sampling-aware** variants of counter/value/histogram processors so the extracted metric remains unbiased by trace sampling decisions. Span-specific extraction is covered in OPIPE-03 (sampling-aware metrics).

Metric Extraction

OpenPipeline's Metric extraction stage has three processors — **Counter**, **Value** and **Histogram** metric. Value and Counter are covered here; Histogram captures a distribution for percentiles.

Value Metrics (Gauge)

Extract numeric values as gauge metrics.

Field	Description
Key	Metric name (e.g., `log.payment.amount`)

Field	Description
Value	Numeric field to extract
Dimensions	Fields to use as metric dimensions
Matching Condition	When to apply extraction

Example: Extract payment amount

```
Metric Type: Value
Key: log.payment.amount
Value: amount
Dimensions: service.name, environment, status
Matching: matchesPhrase(content, "payment") AND isNotNull(amount)
```

Counter Metrics

Count occurrences of matching records.

Field	Description
Key	Metric name (e.g., `log.error.count`)
Dimensions	Fields to use as metric dimensions
Matching Condition	When to count

Example: Count errors

```
Metric Type: Counter
Key: log.errors.count
Dimensions: service.name, error_code
Matching: status == "ERROR"
```

Metric Naming Conventions

```
Format: &lt;source&gt;.&lt;domain&gt;.&lt;name&gt;

Examples:
  log.api.request_count
  log.payment.amount
  log.auth.failed_attempts
  span.checkout.duration_ms
```

Dimension Best Practices

Do	Don't
Low-cardinality fields	High-cardinality IDs
`service.name`	`request_id`
`environment`	`user_id`
`error_code`	`timestamp`

SLI Metric Patterns: RED Methodology ⭐ NEW

The **RED Method** (Rate, Errors, Duration) provides comprehensive service observability.

RED Overview

Metric	Measures	Why
Rate	Requests/sec	Traffic volume, capacity
Errors	Error rate (%)	Reliability, SLA
Duration	Response time (p50, p95, p99)	User experience

Rate: Request Throughput

```
Type: Counter
Key: log.api.request_rate
Dimensions: method, path, status_category, service.name
Matching: isNotNull(path)
```

Errors: Error Rate SLI

Total Requests:

```
Counter: log.api.request_total
Dimensions: service.name, endpoint
```

Error Requests:

```
Counter: log.api.request_errors
Dimensions: service.name, endpoint, error_type
Matching: http.response.status_code &gt;= 500 OR status == "ERROR"
```

> On a log record, `status` is the severity group (`ERROR` , `WARN` , `INFO` , `NONE`), not an HTTP code — parse the HTTP status into its own field, such as `http.response.status_code` , before matching on it.

Duration: Latency Percentiles

```
Type: Value
Key: log.api.response_time_ms
Value: duration_ms
Dimensions: service.name, endpoint, status_category
```

Latency Buckets (Apdex-style SLI)

Use `fieldsAdd` to create latency buckets:

```
fieldsAdd latency_bucket = if(duration_ms &lt; 100, "fast",
                           else: if(duration_ms &lt; 500, "acceptable",
                           else: if(duration_ms &lt; 1000, "slow",
                           else: "very_slow")))
```

Then create a counter metric:

```
Counter: log.api.latency_bucket_count
Dimensions: latency_bucket, service.name, endpoint
```

Event Extraction

Extract events that Dynatrace Intelligence can analyze for root cause analysis.

Davis Event Configuration

Davis events are extracted in the **Davis** stage with the **Davis event** processor.

Field	Description
Event name	Title template — if the event opens a problem, this is the problem title
Event description	Template for the description (placeholders allowed)
Matching condition	When to generate the event
Event properties	Key/value pairs. **`event.type` is required**; everything else is optional

Event Types

`event.type` takes the custom-event type values — the Events API v2 list is `AVAILABILITY_EVENT`, `CUSTOM_ALERT`, `CUSTOM_ANNOTATION`, `CUSTOM_CONFIGURATION`, `CUSTOM_DEPLOYMENT`, `CUSTOM_INFO`, `ERROR_EVENT`, `MARKED_FOR_TERMINATION`, `PERFORMANCE_EVENT`, `RESOURCE_CONTENTION_EVENT` and `WARNING`. Short forms such as `ERROR`, `INFO` or `AVAILABILITY` are **not** event types.

Type	Use Case
`CUSTOM_ALERT`	Alert-worthy condition (the type the OpenPipeline docs use in their example)
`ERROR_EVENT`	Error conditions
`AVAILABILITY_EVENT`	Service up/down events
`RESOURCE_CONTENTION_EVENT`	Resource contention events
`CUSTOM_INFO`	Informational events

On the validation tenant over 7 days (10/02/2026), `dt.davis.events` held `WARNING`, `CUSTOM_INFO`, `AVAILABILITY_EVENT`, `ERROR_EVENT`, `RESOURCE_CONTENTION_EVENT` and `CUSTOM_ALERT` among others.

> **Sources:** Davis stage in OpenPipeline (DT docs) — *"event.type is required; all other properties are optional."*, example `event.type = CUSTOM_ALERT`; Events API v2 - POST an event (DT docs) — `eventType` enumeration.

Event Description Templates

Use field placeholders in descriptions:

```
Payment failed for order {order_id}: {error_message}
Service {service.name} error rate exceeded threshold
Authentication failure for user {user_id} from IP {client_ip}
```

Example Event Extraction

```
Event name: Payment failed for order {order_id}
Event description: Payment failed for order {order_id}: {error_message}
Event properties: event.type = ERROR_EVENT
Matching: matchesPhrase(content, "payment") AND matchesPhrase(content,
"failed")
```

Attribution — what the extracted event attaches to

A Davis event only earns its place if Dynatrace can tell what it is *about*. Most Davis events carry `dt.smartscape_source.id`, the Smartscape entity ID of the signal's source — 84.9% on a validation tenant over 7 days, 08/11/2026 — and the documented grouping rule merges all events sharing that value into one problem. What an extracted event is attributed to when its source record carries no entity is **not documented for OpenPipeline extraction**. The one documented fallback belongs to the Events API v2: an event posted with no `entitySelector` is associated with the environment (`dt.entity.environment`) entity — do not assume extraction behaves the same way.

What a validation tenant showed over 7 days, 09/28/2026: none of its 4,312 OpenPipeline-extracted Davis events carried `dt.smartscape_source.id`; all carried `dt.source_entity` — 4,118 a process group instance, 183 a service, 10 another monitored entity, and **1 the environment entity**. Attribution tracked the entities available on the source records. An event attributed to the environment, or to nothing, has no root cause to point at and nothing to route on.

Extraction reads the record; it cannot resolve an entity that is not on it. A log OneAgent collected from a monitored process carries `dt.entity.host` and `dt.entity.process_group_instance`, and on Kubernetes gains `dt.entity.cloud_application` from entity detection after the Processing stage — in time for the Davis stage. A log pushed through the generic ingest API or forwarded by a third-party shipper frequently arrives with none of that, and no extraction rule can conjure it. Enrichment is therefore a **prerequisite** of event extraction, not a nice-to-have alongside it.

Check the source stream before you configure the extraction, not after:

```
// Entity-enrichment coverage on the stream you plan to extract events from.
// Low coverage here means extracted Davis events will be unattributable.
fetch logs, from:-2h
| summarize total = count(),
    with_host = countIf(isNotNull(dt.entity.host)),
    with_pgi = countIf(isNotNull(dt.entity.process_group_instance)),
    with_workload = countIf(isNotNull(dt.entity.cloud_application))
| fieldsAdd host_pct = round(with_host * 100.0 / total, decimals: 1),
            pgi_pct = round(with_pgi * 100.0 / total, decimals: 1),
            workload_pct = round(with_workload * 100.0 / total, decimals: 1)
```

Executed on a validation tenant over 2 hours, 07/31/2026: **1,369,261 records at 97.8% host, 97.8% process-group, and 97.7% workload coverage** — the shape you expect

where OneAgent collects nearly everything. The residual couple of percent is precisely where an extracted event would land unattributed.

If your own figure is materially lower, in community practice the streams behind the gap are usually API-pushed or forwarder-sourced — verify with the per-source filter below. Fix enrichment there first — OPLOGS-03 §4 covers attribute creation, and its OneAgent attribute-enrichment note covers setting primary fields at source — rather than extracting events that cannot correlate. Filter this query by `dt.openpipeline.source` or `k8s.namespace.name` to isolate the specific stream your extraction rule matches.

Once extraction is live, confirm the events actually carry a key: AIOPS-02 §8 ranks detectors and extracted event sources by firing count and shows which are unattributed, and AIOPS-03 §1 covers the correlation rules themselves.

Business events are a different case entirely — the next section. They never open problems, so none of this applies to them.

> **Sources:** Avoid overalerting (DT docs) — "make sure to set the `dt.smartscape_source.id` field to an existing Smartscape entity ID"; Processing in OpenPipeline (DT docs) — *"Extracts fields into a new record and sends it to an event table."*; Events API v2 - POST an event (DT docs) — *"If not set, the event is associated with the environment"* (scoped to `entitySelector` on that API). **Derived:** treating source-record enrichment as a prerequisite of event extraction follows from the correlation requirement plus the fact that an extraction rule can only read fields present on the record.

Business Event Extraction

Business events are stored in Grail for business analytics.

Business Event Configuration

Field	Description
Event Type	Business event type name
Event Provider	Source system identifier
Data Fields	Fields to include in event

Field	Description
Matching Condition	When to extract

Example: Order Events

```
Event Type: com.example.order.placed
Event Provider: checkout-service
Data Fields:
  - order_id
  - customer_id
  - total_amount
  - currency
  - timestamp
Matching: matchesPhrase(content, "order placed successfully")
```

Querying Business Events

Business KPI Extraction ⭐ NEW

Revenue Tracking

Order Revenue:

```
Type: Value
Key: log.revenue.order_amount
Value: amount
Dimensions: currency, service.name, payment_method
```

Conversion Funnels

E-Commerce Funnel:
1. Product View → 2. Add to Cart → 3. Checkout → 4. Complete

Use `fieldsAdd` to categorize funnel stages:

```
fieldsAdd funnel_stage = if(action == "product_view", "1_view",
                         else: if(action == "add_to_cart", "2_cart",
                         else: if(action == "checkout_started", "3_checkout",
                         else: "4_complete")))
```

Feature Usage

```
Counter: log.feature.usage_count
Dimensions: feature_name, feature_enabled, user_segment
```

```
// Query business events extracted from logs
fetch bizevents, from: now() - 24h
| filter event.type == "com.example.order.placed"
| fields timestamp, order_id, customer_id, total_amount
| limit 50
```

```
// Business event analytics - order totals by hour
fetch bizevents, from: now() - 24h
| filter event.type == "com.example.order.placed"
| makeTimeseries {
    order_count = count(),
    total_revenue = sum(total_amount)
  }, interval: 1h
```

Extraction Matching Conditions

Matching conditions determine which records trigger extraction.

Common Patterns

Pattern	Condition
Error logs	`status == "ERROR"` (also matches `SEVERE`, `CRITICAL`, `FATAL` and the rest; `loglevel == "ERROR"` does not)

Pattern	Condition
Specific service	`service.name == "payment-service"`
Contains text	`matchesPhrase(content, "failed")` (whole token) or `matchesValue(content, "*failed*")` (any substring) — `contains()` is not enabled in matchers
Has field	`isNotNull(order_id)`
Numeric threshold	`duration_ms > 1000`

Combining Conditions

```
// AND - both must match
status == "ERROR" AND service.name == "payment"

// OR - either matches
outcome == "failed" OR outcome == "error"

// Complex conditions
(status == "ERROR" OR status == "WARN")
  AND matchesPhrase(content, "payment")
  AND isNotNull(amount)
```

> **Sources:** Automatic log enrichment (DT docs) — *"Additionally, for each log event, a status attribute is created with a value that is a sum of loglevel values based on the following grouping"*.

Field Reference in Conditions

Use any field available after processing:
- Original fields from ingestion
- Parsed fields from processors
- Computed fields from `fieldsAdd`

Practical Examples

Example 1: API Request Metrics

Source Log:

```
2024-12-12T10:30:45 INFO - GET /api/users completed in 150ms, status=200
```

Pipeline Processing:

```
// First, parse the log
// LD? skips the timestamp and level; without the ' - ' anchor, LD:method
would capture the timestamp
parse content, "LD? ' - ' LD:method ' ' LD:path ' completed in '
INT:duration_ms 'ms, status=' INT:status_code"
```

Metric Extraction 1: Request duration

```
Type: Value
Key: log.api.request_duration_ms
Value: duration_ms
Dimensions: method, path, status_code
Matching: isNotNull(duration_ms)
```

Metric Extraction 2: Request count

```
Type: Counter
Key: log.api.request_count
Dimensions: method, path, status_code
Matching: isNotNull(path)
```

Example 2: Error Event Generation

Source Log:

```
2024-12-12T10:30:45 ERROR PaymentService - Transaction failed: insufficient
funds, orderId=12345
```

Event Extraction:

```
Event name: Transaction failed for order {order_id}
Event description: Transaction failed for order {order_id}: {error_reason}
Event properties: event.type = ERROR_EVENT
Matching: status == "ERROR" AND matchesPhrase(content, "Transaction failed")
```

Example 3: Login Business Events

Source Log:

```
2024-12-12T10:30:45 INFO AuthService - User login successful: userId=john123, ip=192.168.1.100
```

Business Event Extraction:

```
Event Type: com.example.auth.login
Event Provider: auth-service
Data Fields: user_id, client_ip, timestamp
Matching: matchesPhrase(content, "login successful")
```

Example 4: Combined Extraction

From a single payment log line, extract:

1. **Metric**: `log.payment.amount` (value: amount)
2. **Metric**: `log.payment.count` (counter)
3. **Event**: Payment processed (`CUSTOM_INFO`)
4. **Business Event**: Order transaction

Validating Extractions

After configuring extractions, verify they're working.

List log-extracted metrics: browse them in the Dynatrace UI (**Observe > Metrics**), or query a specific metric with `timeseries avg_value = avg(log.your_metric_name), from: now() - 24h` .

```
// Query a specific extracted metric
// Replace 'log.api.request_duration_ms' with your metric key
timeseries {
    avg_duration = avg(log.api.request_duration_ms),
    max_duration = max(log.api.request_duration_ms)
  }, from: now() - 2h, interval: 5m
```

```
// View extracted counter metric by dimension
// Replace 'log.api.request_count' with your metric key
timeseries {
    requests = sum(log.api.request_count)
  }, from: now() - 2h, interval: 5m
```

```
// View extracted events
fetch events, from: now() - 24h
| filter isNotNull(event.name)
| summarize {event_count = count()}, by: {event.type, event.name}
| sort event_count desc
```

```
// View specific event details
fetch events, from: now() - 24h
| filter matchesValue(event.name, "Transaction failed for order*")
| fields timestamp, event.name, event.description, event.type
| sort timestamp desc
| limit 25
```

```
// View business events by type
fetch bizevents, from: now() - 24h
| summarize {bizevent_count = count()}, by: {event.type, event.provider}
| sort bizevent_count desc
```

```
// Compare source log volume to extracted metrics
// This helps verify extraction is capturing expected data
fetch logs, from: now() - 1h
| filter contains(content, "payment")
| summarize {log_count = count()}
| fieldsAdd data_source = "logs"

// Then compare with:
// timeseries sum(log.payment.count)
```

Metric Design Best Practices ⭐ NEW

Cardinality Management

Cardinality	Examples	Max Values	Use?
Low	environment, region	< 50	✅ Ideal
Medium	endpoint, error_code	50-500	✅ Good
High	session_id, request_id	500-10K	⚠️ Avoid
Very High	user_id, order_id	> 10K	❌ Never

Cardinality Example:

```
✅ GOOD: 5 services × 3 environments × 4 methods = 60 time series
❌ BAD: 5 services × 100K user_ids = 500K time series
```

Reduction Techniques:

```
// Bucket high-cardinality values
fieldsAdd duration_bucket = if(duration_ms &lt; 100, "&lt;100ms",
                           else: if(duration_ms &lt; 500, "100-500ms",
                           else: "&gt;500ms"))

// Use segments instead of IDs
fieldsAdd user_segment = if(user_tier == "premium", "premium", else: "free")
```

Naming Conventions

Format: `<source>.<domain>.<metric>.<unit>`

Examples:

```
✅ log.api.request_count
✅ log.api.response_time_ms
✅ log.payment.order_amount_usd
✅ log.auth.failed_login_count

❌ apiReqCnt (abbreviated, no unit)
❌ response_time (no source, no unit)
❌ count (no context)
```

Cost Optimization

Metrics vs. Logs:

```
Scenario: 1M requests/day

Option 1: Store logs (35 days)
  - 35M log records × 500 bytes = 17.5 GB
  - Cost: ~$140/month (placeholder rate — use your rate card)

Option 2: Extract metrics + No storage assignment for the logs
  - 10 time series, kept for the metrics bucket's retention (15 months by
default) = ~10 MB
  - Cost: ~$1/month (placeholder rate — use your rate card)
  - Savings: 99.3% 🎉
```

The dollar figures are illustrative placeholders, not Dynatrace prices; the ratio (17.5 GB of logs against ~10 MB of metrics) is the point. Price it with your own rate card — see FINOPS-01.

When to Extract:
- ✅ SLI tracking (RED)
- ✅ Business KPIs
- ✅ High-volume health checks
- ❌ Debugging (keep logs)
- ❌ Compliance/audit (keep logs)

Best Practices

Metric Extraction

Practice	Reason
Use meaningful metric names	Easier discovery and querying
Limit dimensions (≤5-7)	Avoid metric cardinality explosion
Use low-cardinality dimensions	IDs cause metric bloat

Practice	Reason
Extract only what you need	Reduces storage costs
Test matching conditions	Ensure correct data selection

Event Extraction

Practice	Reason
Meaningful event names	Easier to find in Dynatrace Intelligence
Descriptive templates	Context for troubleshooting
Appropriate event types	Correct Dynatrace Intelligence handling
Avoid over-extraction	Too many events = noise

Business Event Extraction

Practice	Reason
Consistent event types	Enables analytics
Include all needed fields	Avoids re-processing
Use namespaced types	Prevents collisions
Document event schema	Helps consumers

Cost Optimization Pattern

Extract → Don't store → Save

1. Extract metrics and events from verbose logs
2. Assign the original verbose logs **No storage assignment** (Bucket assignment stage) — the record still runs through the extraction stages but is not stored
3. Keep only derived signals

Do not use a **Drop record** processor for step 2: it runs in the Processing stage, before any extraction stage, so nothing would be extracted.

> **Sources:** Processing in OpenPipeline (DT docs) — Drop record: *"Drops a record. The record isn't processed further and isn't stored."*; No storage assignment: *"The record continues through all configured pipeline stages and isn't stored only at the end of the pipeline."*

This pattern reduces storage while preserving observability.

Complete Extraction Pipeline Example

Pipeline: `checkout-service-logs`

Processing Stage (from OPMIG-06):

```
// Parse order details
// LD? lets each key sit anywhere in the line; NSPACE stops at the next space
parse content, "LD? 'orderId=' INT:order_id"
| parse content, "LD? 'amount=' DOUBLE:amount"
| parse content, "LD? 'status=' NSPACE:order_status"
| parse content, "LD? 'duration=' INT:duration_ms 'ms'"
```

Metric Extraction 1: Order Amount

```
Type: Value
Key: log.checkout.order_amount
Value: amount
Dimensions: order_status, service.name
Matching: isNotNull(amount)
```

Metric Extraction 2: Order Count

```
Type: Counter
Key: log.checkout.order_count
Dimensions: order_status, service.name
Matching: isNotNull(order_id)
```

Metric Extraction 3: Processing Duration

```
Type: Value
Key: log.checkout.processing_duration_ms
Value: duration_ms
Dimensions: order_status, service.name
Matching: isNotNull(duration_ms)
```

Event Extraction: Order Failure

```
Event name: Order {order_id} failed
Event description: Order {order_id} failed with status {order_status}
Event properties: event.type = ERROR_EVENT
Matching: order_status == "failed" OR order_status == "error"
```

Business Event: Order Completed

```
Event Type: com.example.checkout.order_completed
Event Provider: checkout-service
Data Fields: order_id, amount, order_status, duration_ms
Matching: order_status == "completed" OR order_status == "success"
```

Next Steps

Now that you can extract metrics and events, continue with:

Notebook	Focus Area
OPMIG-08	Security, Masking & Compliance
OPMIG-09	Troubleshooting & Validation

References

- OpenPipeline Data Extraction
- Extract metrics from spans and distributed traces (DT docs)
- OpenPipeline use cases (DT docs)
- Dynatrace Intelligence Events

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.